INCH-POUND

MIL-DTL-16878/37A
11 August 2000
SUPERSEDING
MIL-W-16878/37(NAVY)
11 September 1992

DETAIL SPECIFICATION SHEET

# WIRE, ELECTRICAL, ETHYLENE-PROPYLENE DIENE ELASTOMER (EPDM) INSULATED 125 °C, 600 VOLTS, EXTRUDED INSULATION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-16878G.

REQUIREMENTS.

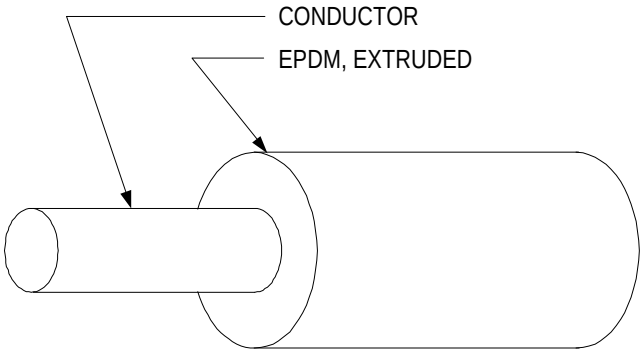


FIGURE 1. Wire configuration.


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TABLE I. Wire configuration and dimensions.

| PIN 1/ | Wire size | Stranding | Conductor | | Conductor Diameter (nominal) (inch) | Finished wire diameter (inch) | |
|---|---|---|---|---|---|---|---|
| | | | Material | Coating | | Min | Max |
| M16878/37BHA* | 18 | 1 X 18 | Copper | Tin | .0403 | .080 | .095 |
| M16878/37BHB* | 18 | 7 X 26 | Copper | Tin | .0480 | .090 | .100 |
| M16878/37BHE* | 18 | 19 X 30 | Copper | Tin | .0500 | .090 | .100 |
| M16878/37BJA* | 16 | 1 X 16 | Copper | Tin | .0508 | .090 | .100 |
| M16878/37BJE* | 16 | 19 X 29 | Copper | Tin | .0570 | .097 | .112 |
| M16878/37BKE* | 14 | 13 X 27 | Copper | Tin | .0720 | .112 | .127 |
| M16878/37BLE* | 12 | 19 X 25 | Copper | Tin | .0910 | .131 | .146 |
| M16878/37BLG* | 12 | 37 X 28 | Copper | Tin | .0890 | .129 | .143 |
| M16878/37BMG* | 10 | 37 X 26 | Copper | Tin | .1110 | .151 | .165 |
| M16878/37BNL* | 8 | 133 X 29 | Copper | Tin | .1690 | .224 | .235 |
| M16878/37BPL* | 6 | 133 X 27 | Copper | Tin | .2130 | .268 | .280 |
| M16878/37BRL* | 4 | 133 X 25 | Copper | Tin | .2690 | .324 | .334 |
| M16878/37BSP* | 2 | 665 X 30 | Copper | Tin | .3150 | .370 | .380 |
| M16878/37BTR* | 1 | 817 X 30 | Copper | Tin | .3850 | .460 | .475 |
| M16878/37BUS* | 0 | 1045 X 30 | Copper | Tin | .4100 | .485 | .495 |
| M16878/37BWT* | 00 | 1330 X 30 | Copper | Tin | .4550 | .530 | .550 |
| M16878/37BYV* | 000 | 1672 X 30 | Copper | Tin | .5350 | .610 | .650 |
| M16878/37BZW* | 0000 | 2109 X 30 | Copper | Tin | .5950 | .670 | .800 |

Notes:
1/ PIN stands for part or identifying number.

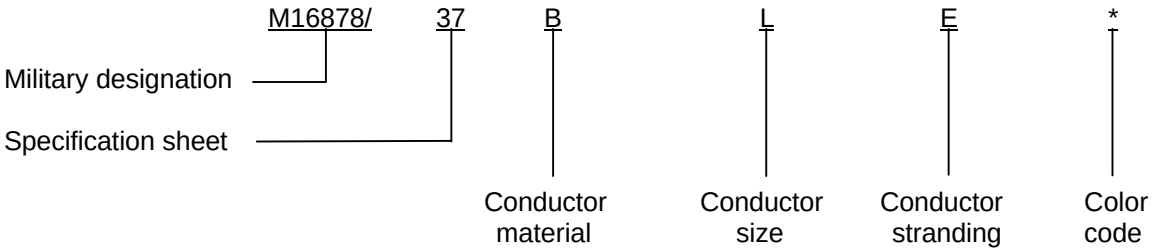


FIGURE 2. Example of PIN (see MIL-DTL-16878G).

| | |
|---|---|
| Configuration and dimensions: | See figure 1 and table I |
| Operating voltage: | Up to 600 volts |
| Operating temperature: | Up to 125 °C |
| Insulation: | Extruded ethylene-propylene diene elastomer (EPDM) |
| Spark test voltage: | 6.0 kV |
| Impulse dielectric test voltage: | 6.7 kV to 7.8 kV, or 9.5 to 11.0 kV using the 3.0 kHz spark test |
| Dielectric withstanding voltage: | 5.0 kV |



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| | |
|---|---|
| Insulation resistance: | $IR = K \log_{10} D/d$ |
| | Where: IR = Minimum insulation resistance in megohms per 1000 feet at 20 °C |
| | K = 20,000 |
| | D = Maximum average diameter of finished wire |
| | d = Conductor diameter |
| Cold bend: | Condition 4 hours at -25±1 °C (see table II) |

TABLE II. Cold bend mandrel sizes.

| Wire size | Cold bend mandrel diameter (inches, maximum) |
|---|---|
| 18 through 10 | .5 |
| 8 through 2 | 1.5 |
| 1 through 0000 | 4.0 |

| | |
|---|---|
| Surface resistance: | Not required |
| Heat resistance: | Condition at 150 °C |
| Heat aging: | Not required |
| Insulation tensile strength: | 1000 pounds force per square inch (minimum) |
| Insulation elongation: | 250 percent (minimum) |
| Identification of product: | Required |

CHANGES FROM PREVIOUS ISSUE. Marginal notations are not used in this revision to identify changes with respect to the previous issue because of the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Navy - SH
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 6145-2193-020)

Review activity:
Navy - AS



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